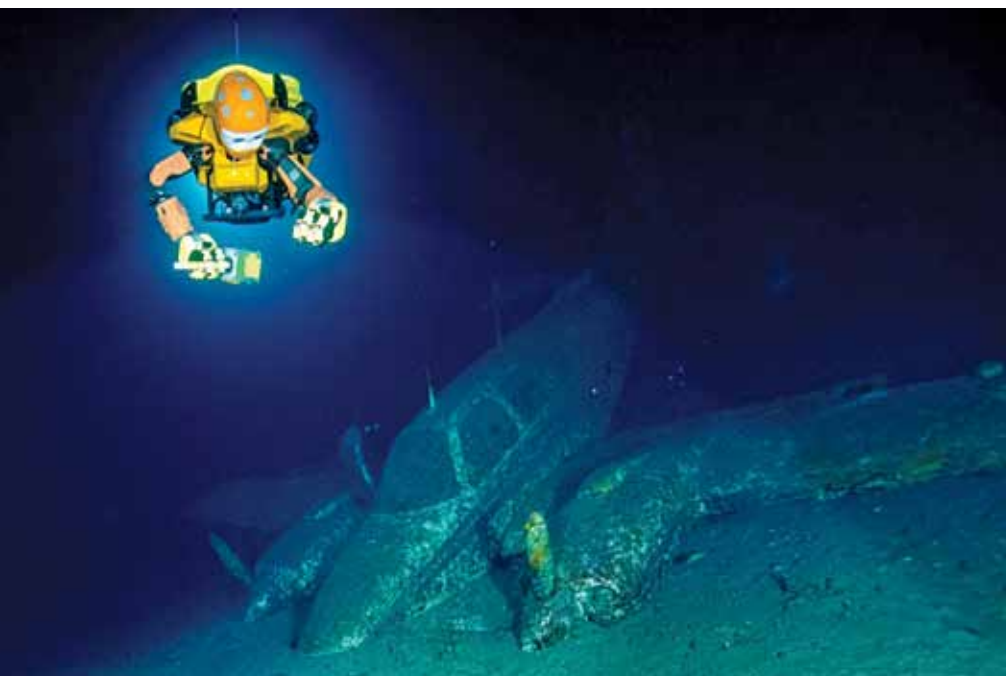


Stanford's underwater robot goes on expedition



OceanOneK with its boom camera in hand near a Beechcraft Baron F-GDPV aircraft that was 67m down (over 200 ft) (Credit: <https://news.stanford.edu>)

Stanford University roboticists have created an underwater robot, OceanOneK, and successfully sent it on an exciting expedition! OceanOneK's unique characteristics give its users the impression that they are actually visiting these deep-water locations. It can carefully manoeuvre underwater thanks to its eight multi-directional thrusters on its thinner back half and humanoid upper half. The haptic or touch based feedback system of the robot and stereoscopic vision provide astonishingly lifelike feelings that are equivalent to what the user would have felt if she/he had been below the surface instead of above it on the control ship.

New radar communicates, navigates, undertakes e-warfare

Most military-grade systems are still analogue in nature. Radars are now getting a significant digital upgrade from Sandia National Laboratories. Sandia's new digital radar can be altered to perform various tasks, including communication, navigation, and electronic warfare. On equipment approximately the size of a small toolbox, users will be able to download the software and firmware they require for each mission. The new digital architecture may run numerous radio-frequency channels concurrently, either for a single function or operating independently on a number of separate functions thanks to advanced wireless technology.



Sandia National Laboratories principal investigator Jacques Loui, left, and a firmware developer are part of a team redesigning high-performance radar as a flexible, multipurpose sensor (Credit: <https://www.eurekalert.org>)